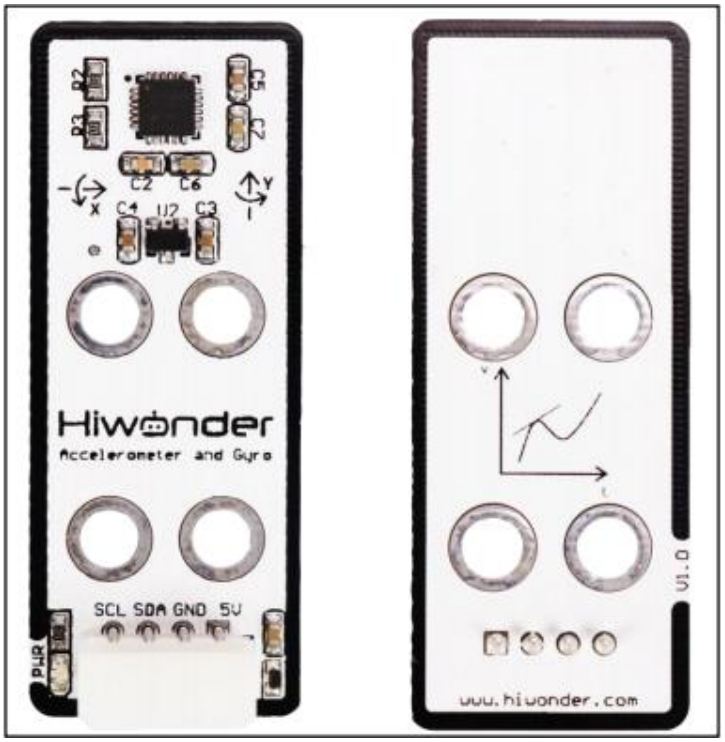
Shenzhen Hiwonder Technology Co,Ltd

Hiwonder Acceleration Sensor

V1.0



1. Acceleration Sensor Instruction

1.1 Acceleration Sensor Introduction

Hiwonder acceleration sensor is widely used, and it is mainly applicable in handheld game products, 3D remote controls, and portable navigation, etc.

1.2 Working Principle

This sensor relies on the MPU6050 sensor component, which integrates a 3-axis MEMS gyroscope, a 3-axis MEMS accelerometer, and a configurable Digital Motion Processor (DMP).

Utilizing three 16-bit ADCs, the sensor converts analog measurements from the gyroscope and accelerometer into digital output. The sensor's measurement ranges are adjustable by the user to accurately track both fast and slow movements. The gyroscope's measurable range spans ±250, ±500, ±1000, ±2000 degrees per second (dps), while the accelerometer's range is ±2, ±4, ±8, ±16g.

2. Precaution

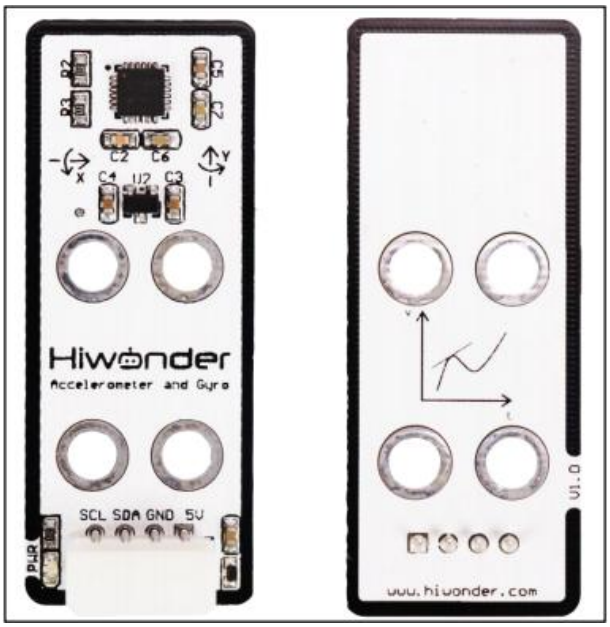
① Do not use the acceleration sensor beyond its rated voltage range (5V).

② Please handle it with care during your usage to avoid dropping or damaging.

3. Specification Instruction

Please access “3. Chip Manual and Schematic Diagram” to view the chip introduction of the acceleration sensor.

3.1 Pin Instruction



PIN	PIN Instruction
5V	Power Supply
GND	GND
SDA	SDA

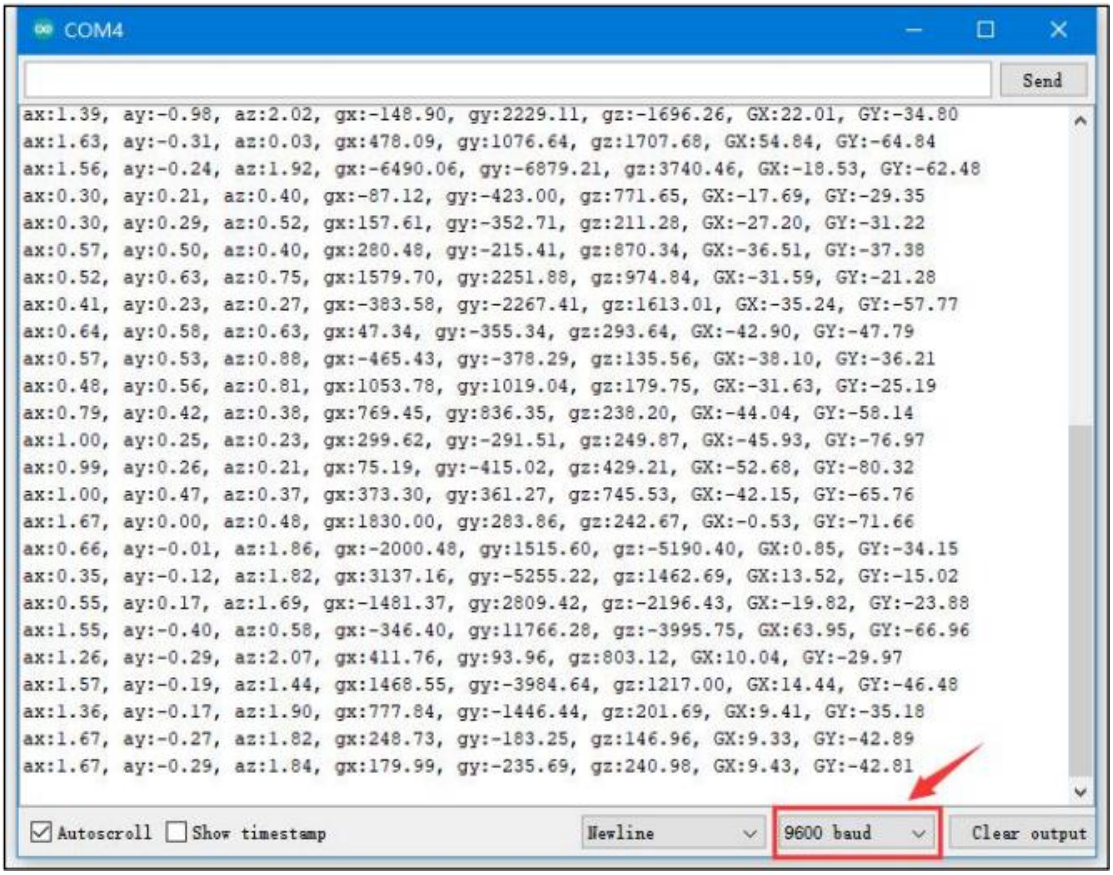
SCL	SCL

3.2 Specification

Acceleration Sensor	
Name	Specification
Working Voltage	DC 5V
Data Port	I2C. The fastest clock rate of I2C can reach 400kHz.
Port Type	5264-4AW
Accelerometer Measurable Range	±2，±4，±8，±16g
PWR Instruction	Lighting up after powering.
Size	50mmx20mm
Easy to install, compatible with Lego series.	

4. Program Outcome

Please refer to the case tutorials and programs on various platforms in the same directory as this tutorial. In this tutorial, Arduino IDE is used as an example to demonstrate the project outcome.

Use the MPU6050 sensor element to measure the acceleration in three

axes. Convert the measured analog quantity into digital quantity and output it via the I2C bus. When you randomly rotate the acceleration sensor, you can see the acceleration values in a certain direction change on the serial monitor.